Turbulent Pipe-Flow Model Comparison

Aim: To compare radial eddy-viscosity closures using mean velocity, friction factor, pressure drop, and turbulent viscosity.

Case definition and parameters: Simulations were analyzed on the following parameters.

Parameter	Value
vtk, nx	160
vtk, ny	61
Re_min	10^4, 10^5
Re_max	10^4,10^5
N_Re	3
Max iterations	1500
Tolerance	1e-07

Governing Equation:

For fully developed axisymmetric mean flow,

$$\frac{1}{r}\frac{d}{dr}\left[r(\nu+\nu_t)\frac{dU}{dr}\right]=\frac{1}{\rho}\frac{dp}{dx}.$$

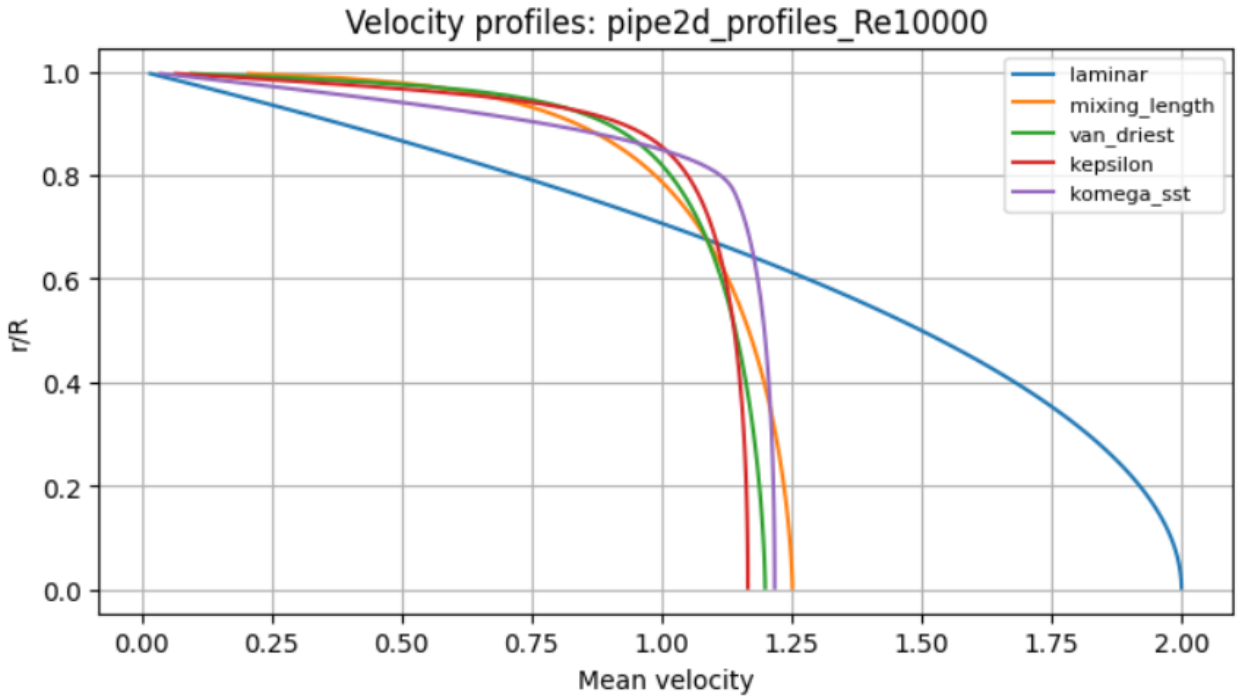
The closures differ through the model for νt

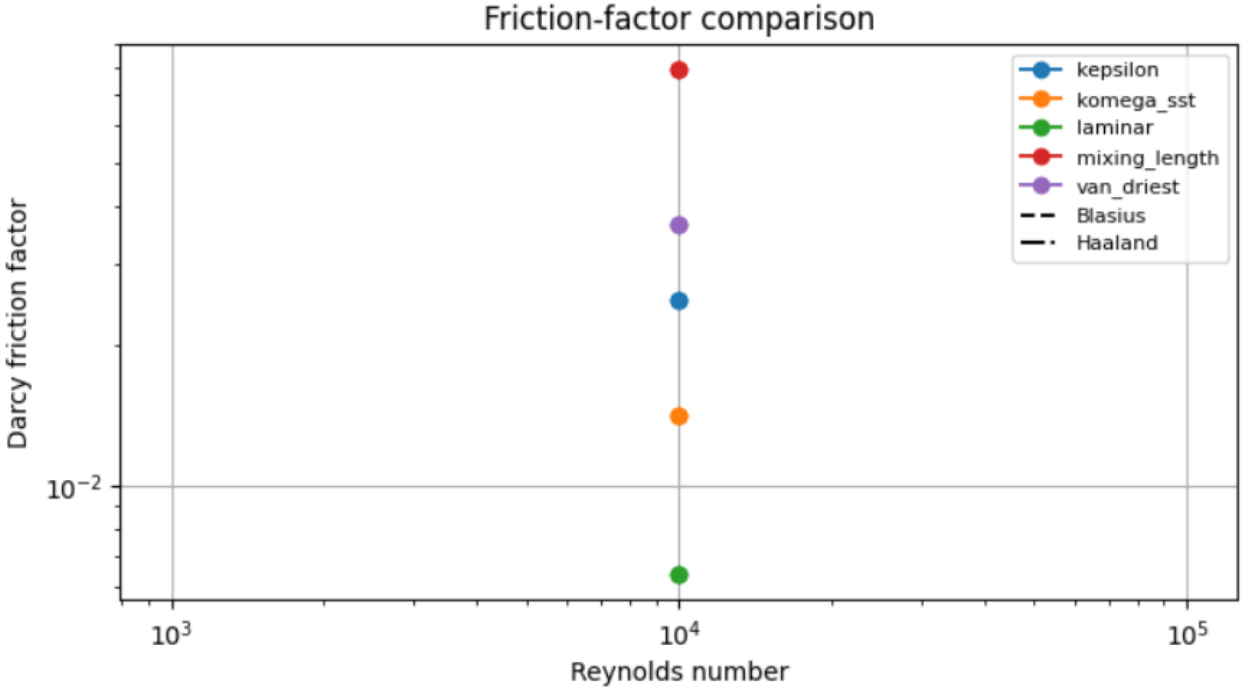
Boundary Conditions: Axis symmetry at the centreline, no-slip at the wall, and pressure gradient adjusted to obtain unit bulk velocity.

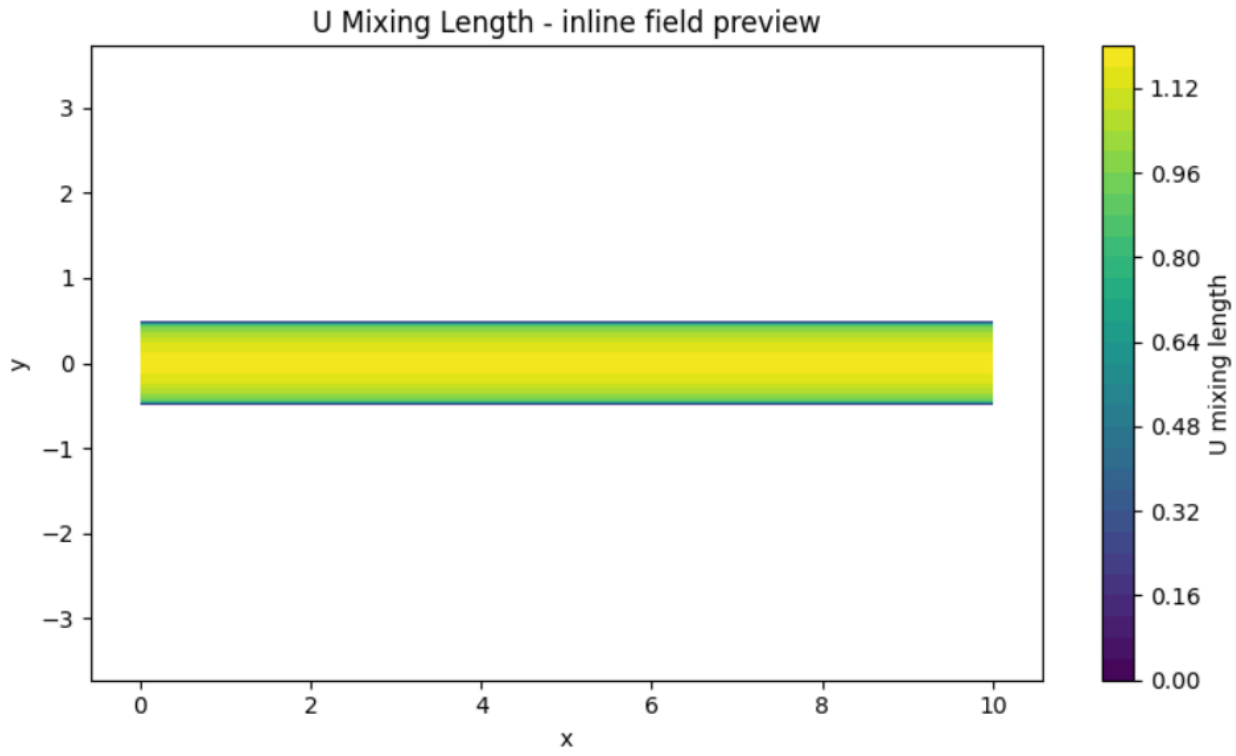
Numerical Method: Radial finite-volume momentum solution for laminar, mixing-length, Van-Driest-damped, equilibrium k-ε, and equilibrium k-ω/SST surrogate closures.

Observations: Velocity profile fullness, friction factor, Blasius and Haaland references, wall shear, pressure drop, and maximum turbulent-viscosity ratio at Re = 10^4, approximately 3.16×10^4, and 10^5. Explain which closure gives the fullest profile and compare friction factors with reference correlations.

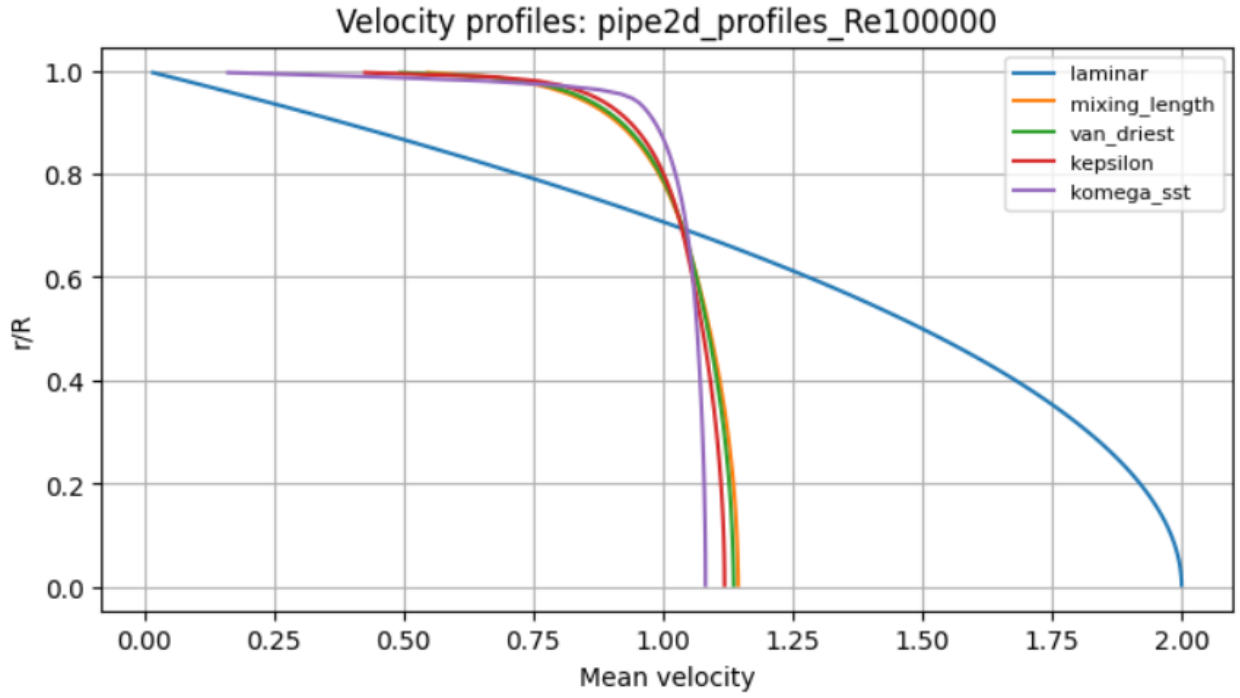
For Raynolds number 10^4:

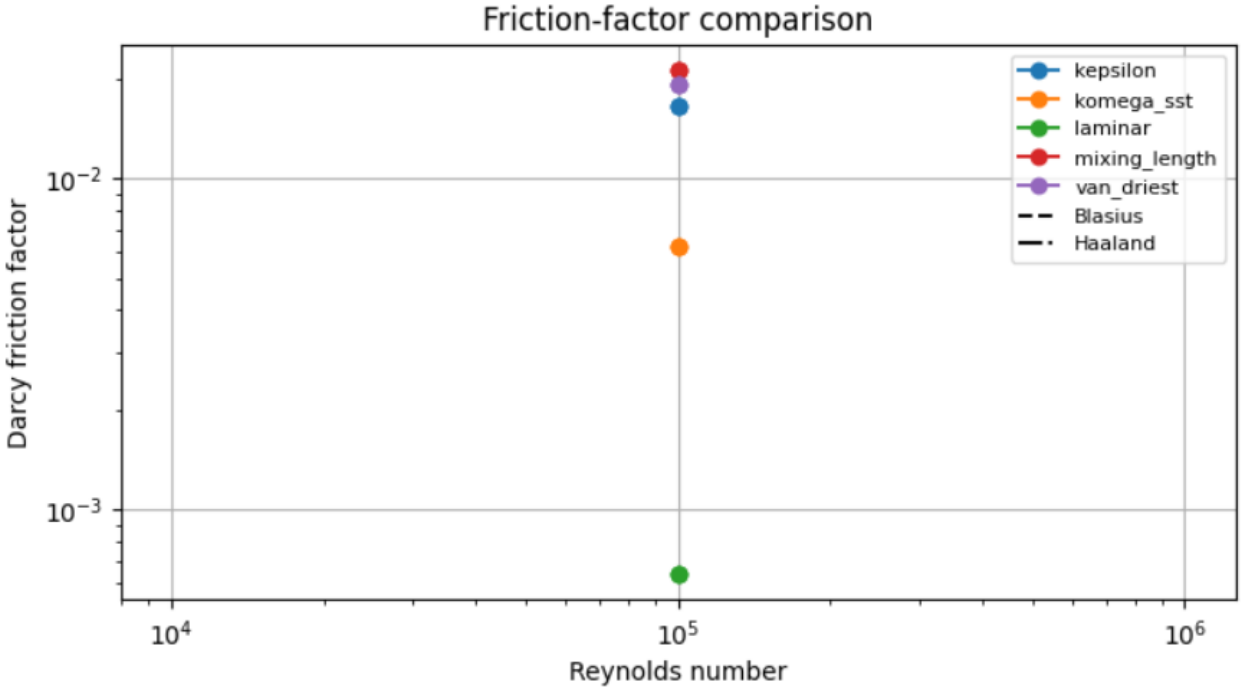

Velocity profiles: pipe2d_profiles_Re10000
laminar
mixing_length
van_driest
kepsilon
komega_sst
r/R
Mean velocity
0.0
0.2
0.4
0.6
0.8
1.0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00

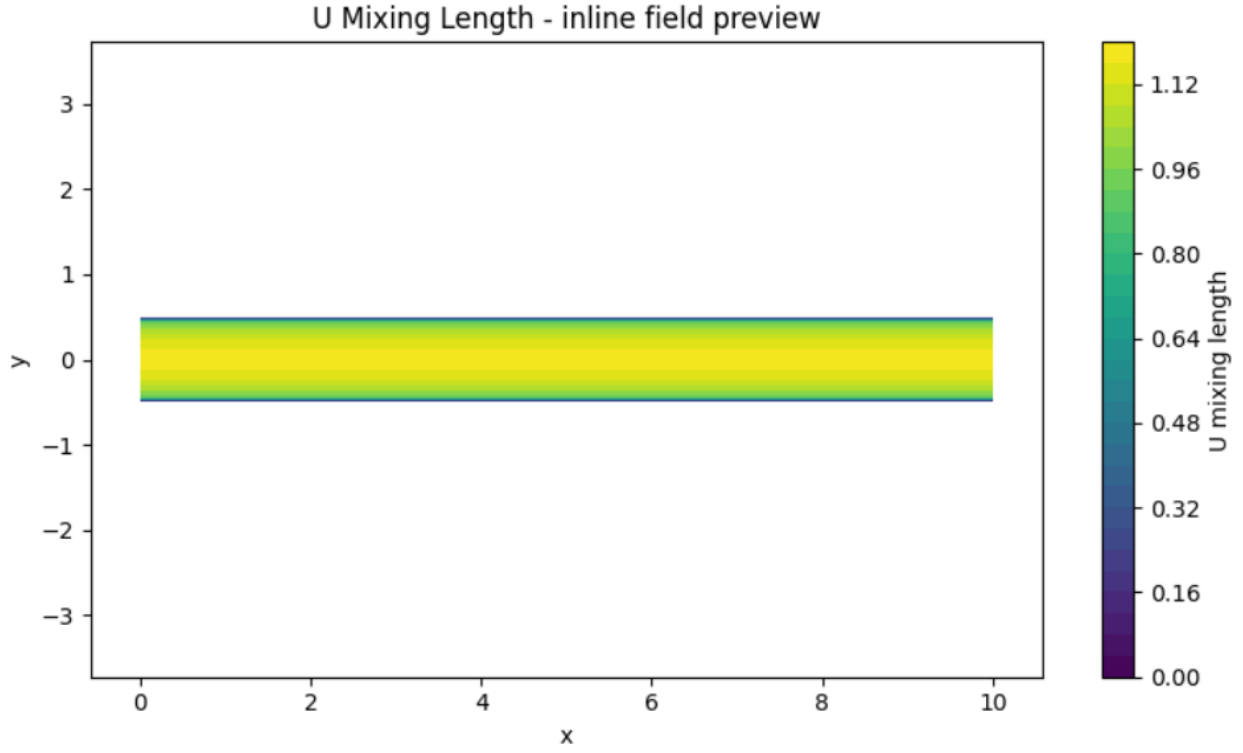

Friction-factor comparison
kepsilon
komega_sst
laminar
mixing_length
van_driest
Blasius
Haaland
Darcy friction factor
Reynolds number
10^{-2}
10^{3}
10^{4}
10^{5}



For Raynolds number 10^5:




Friction-factor comparison
kepsilon
komega_sst
laminar
mixing_length
van_driest
Blasius
Haaland
Darcy friction factor
10^{-2}
10^{-3}
10^{4}
10^{5}
10^{6}
Reynolds number


U Mixing Length - inline field preview
y
x
U mixing length
1.12
0.96
0.80
0.64
0.48
0.32
0.16
0.00

Results:

Parameter	**Re = 10^4**	**Re = 10^5**
Blasius friction factor (reference)	0.0316	0.0178
Haaland friction factor, smooth pipe (reference)	0.0309	0.0178
Centreline velocity, U_cl (fullness U_b / U_cl): laminar	2.00 (0.50)	2.00 (0.50)
Mixing length	≈ 1.25 (0.80)	≈ 1.14 (0.88)
Van Driest	≈ 1.20 (0.83)	≈ 1.14 (0.88)
k–ε	≈ 1.165 (0.86)	≈ 1.115 (0.90)
k–ω / SST surrogate	≈ 1.20 (0.83)	≈ 1.08 (0.93)
Darcy friction factor (ratio to Blasius): laminar	≈ 0.0064 (0.20)	≈ 0.00064 (0.04)
Mixing length	≈ 0.083 (2.6)	≈ 0.021 (1.20)
Van Driest	≈ 0.038 (1.19)	≈ 0.019 (1.09)
k–ε	≈ 0.026 (0.81)	≈ 0.0165 (0.93)
k–ω / SST surrogate	≈ 0.014 (0.45)	≈ 0.0062 (0.35)
Wall shear and pressure drop	proportional to f at fixed U_b	proportional to f at fixed U_b

Conclusion:

Radial eddy-viscosity closures (laminar, mixing-length, Van-Driest-damped mixing-length, equilibrium k–ε and a k–ω/SST surrogate) were compared for fully developed turbulent pipe flow at a unit bulk velocity, with the pressure gradient adjusted to obtain that bulk velocity, at Re = 10^4 and 10^5. The laminar case recovers the exact Poiseuille solution, with a centreline velocity of 2.00 and a friction factor of 64/Re (≈ 0.0064 and ≈ 0.00064), which confirms that the momentum solver

and the friction-factor evaluation are correct. It is included as a reference, since the flow is not laminar at these Reynolds numbers.

Profile fullness: All turbulent closures give much fuller profiles than the parabolic laminar one, with a flat core and a steep drop in a thin layer near the wall. Using the ratio of bulk to centreline velocity as a measure of fullness (0.5 for laminar), the values at $Re = 10^4$ are about 0.80 (mixing length), 0.83 (Van Driest), 0.86 (k–ε) and 0.83 (k–ω/SST surrogate), so k–ε gives the fullest profile by this measure, although the k–ω profile has the flattest core and the sharpest near-wall drop. At $Re = 10^5$ the ratios are about 0.88, 0.88, 0.90 and 0.93, and the k–ω/SST surrogate gives the fullest profile. Profiles become fuller as Re increases for every closure. The mixing-length and Van Driest profiles are almost identical at $Re = 10^5$, because wall damping only affects a thin viscous layer that shrinks as Re rises. For comparison, the usual relation $U_{cl}/U_b \approx 1 + 1.43\sqrt{f}$ gives about 1.25 at $Re = 10^4$ and 1.19 at $Re = 10^5$ for smooth pipes. The mixing-length result at $Re = 10^4$ (≈ 1.25) matches this, while all other closures are slightly fuller than expected, and at $Re = 10^5$ all closures are fuller (1.08–1.14).

Friction factor against the references: The Blasius correlation gives 0.0316 and 0.0178, and the Haaland correlation 0.0309 and 0.0178, at $Re = 10^4$ and 10^5. The Van Driest closure is closest among the high-friction models, at about 19% above Blasius at $Re = 10^4$ and 9% above at $Re = 10^5$. The k–ε closure is slightly below the correlations (about 0.81 and 0.93 times Blasius). The undamped mixing-length closure over-predicts friction strongly, by a factor of about 2.6 at $Re = 10^4$ and 1.2 at $Re = 10^5$. The k–ω/SST surrogate under-predicts it severely, at only about 0.45 and 0.35 times Blasius, which is not physically acceptable for smooth-pipe flow. Because wall shear and pressure drop at fixed bulk velocity are both proportional to the friction factor, they follow the same ordering and the same errors.

Consistency of the results: The friction factor and the profile fullness do not agree for every closure. A mixing-length friction factor of 0.083 at $Re = 10^4$ would imply a centreline velocity of about 1.4 through the relation above, but the computed profile has about 1.25. Likewise, the k–ω/SST surrogate gives the fullest profile at $Re = 10^5$ but the lowest friction. This suggests that the wall shear used for the friction factor is sensitive to the near-wall treatment, for example the first-cell velocity gradient or a wall function, and not only to the bulk profile. The velocity profiles of the k–ε and k–ω closures also end at non-zero velocity at $r/R = 1$, which suggests that their first point lies outside the viscous sublayer, as in a wall-function approach. The friction values from these two closures should therefore be treated as less reliable than the profile shapes.

Limitations of the figures: The friction-factor plots show one Reynolds number each, and the Blasius and Haaland lines are listed in the legends but not drawn, so the comparison above uses the correlation values calculated from the formulas. The intermediate case at $Re \approx 3.16 \times 10^4$, the wall shear, the pressure drop and the maximum turbulent-viscosity ratio were requested in the observations but are not shown in the figures. The field plot is available only for the mixing-length

case at Re = 10^4, where the streamwise velocity is uniform along the pipe with a maximum of about 1.2, as expected for a fully developed flow.

Overall, every turbulent closure produces the expected fuller-than-laminar profile, and the profiles become fuller with Reynolds number. The Van Driest mixing-length model gives friction factors closest to the Blasius and Haaland correlations (within about 20%), k–ε is slightly low, the plain mixing-length model is too high at Re = 10^4, and the k–ω/SST surrogate under-predicts the friction by more than half. Fullness alone is therefore not a measure of correctness: the closure with the fullest profile at Re = 10^5 also has the least accurate friction factor. Plotting the correlations, showing the Re ≈ 3.16×10^4 case, checking the near-wall resolution and wall treatment (y^+ of the first cell), and reporting the turbulent-viscosity ratio would complete the comparison.